PHASE 2: SQL DATA CLEANING

1. Remove duplicates (DISTINCT)

5 minutes ago (1s)

15: Remove duplicates

```
SELECT DISTINCT *
FROM insurance_data_clean
```

See performance (1)

Optimize

Table

	Customer_ID	Age	Gender	Province	Monthly_Income	Join_Date	Policy_ID	Policy_Type	Premium_Amount	Policy_S
1	CUST1000	44	Male	Gauteng	26064	2020-05-30	POL5000	Home	1495	Active
2	CUST1001	41	Male	Free State	3500	2025-06-16	POL5001	Auto	746.28	Active
3	CUST1002	33	Female	Limpopo	29552.92	2020-06-14	POL5002	Health	2325.42	Active
4	CUST1003	41	Female	western cape	3500	2024-06-10	POL5003	Auto	885.13	Active
5	CUST1004	42	Female	Mpumalanga	3500	2024-07-15	POL5004	Home	1875.25	Active
6	CUST1005	33	Female	Limpopo	null	2024-04-13	POL5005	Home	2210.31	Active
7	CUST1006	38	Male	Free State	49905.01	2022-12-19	POL5006	Motor	433.09	Active
8	CUST1007	30	FEMALE	KwaZulu-Natal	20975.4	null	POL5007	Life	817.56	Active
9	CUST1008	42	Male	North West	31654.77	2022-03-28	POL5008	Home	1410.71	Active
10	CUST1009	41	Male	Limpopo	26957.76	2020-11-02	POL5009	Funeral	326.92	Active
11	CUST1010	47	Female	Western Cape	15268.58	2021-05-18	POL5010	Motor	1840.98	Active
12	CUST1011	27	Female	North West	34630.29	null	POL5011	Auto	1028.2	Cancelled
13	CUST1012	41	Male	Western Cape	38945.52	2021-08-24	POL5012	Home	1734	Active
14	CUST1013	59	Female	Gauteng	34633.66	2021-01-27	POL5013	Life	706.33	Active
15										

538 rows | 0.95s runtime

Refreshed 4 minutes ago

2. Handle NULL values (COALESCE)

09:44 PM (4s) 16: Handle NULL values (COALESCE)

```sql
CREATE OR REPLACE TABLE insurance_data_clean AS
SELECT
  COALESCE(NULLIF(TRIM(Customer_ID), ''), 'Unknown') AS Customer_ID,
  COALESCE(Age, 0) AS Age,
  COALESCE(NULLIF(TRIM(Gender), ''), 'Unknown') AS Gender,
  COALESCE(NULLIF(TRIM(Province), ''), 'Unknown') AS Province,

  -- Numeric cleaning
  COALESCE(
    TRY_CAST(NULLIF(REGEXP_REPLACE(Monthly_Income, '[R,\\s]', ''), '') AS DOUBLE),
    0
  ) AS Monthly_Income,

  -- Join_Date with garbage handling
  CASE
    WHEN Join_Date IS NULL OR TRIM(Join_Date) IN ('', 'N/A', 'UNKNOWN', '--')
      THEN DATE('1900-01-01')
    ELSE COALESCE(
      TRY_TO_DATE(Join_Date, 'yyyy-MM-dd'),
      TRY_TO_DATE(Join_Date, 'dd-MMM-yyyy'),
      TRY_TO_DATE(Join_Date, 'MM/dd/yyyy'),
      TRY_TO_DATE(Join_Date, 'dd/MM/yyyy'),
      DATE('1900-01-01')
    )
  END AS Join_Date,

  COALESCE(NULLIF(TRIM(Policy_ID), ''), 'Unknown') AS Policy_ID,
  COALESCE(NULLIF(TRIM(Policy_Type), ''), 'Unknown') AS Policy_Type,

  COALESCE(
    TRY_CAST(NULLIF(REGEXP_REPLACE(Premium_Amount, '[R,\\s]', ''), '') AS DOUBLE),
    0
  ) AS Premium_Amount,

  COALESCE(NULLIF(TRIM(Policy_Status), ''), 'UNKNOWN') AS Policy_Status,
```

```sql
  -- Claim_ID with garbage handling
  COALESCE(NULLIF(TRIM(Claim_ID), ''), 'Unknown') AS Claim_ID,

  -- Claim_Date with garbage handling
  CASE
    WHEN Claim_Date IS NULL OR TRIM(Claim_Date) IN ('', 'N/A', 'UNKNOWN', '--')
      THEN DATE('1900-01-01')
    ELSE COALESCE(
      TRY_TO_DATE(Claim_Date, 'yyyy-MM-dd'),
      TRY_TO_DATE(Claim_Date, 'dd-MMM-yyyy'),
      TRY_TO_DATE(Claim_Date, 'MM/dd/yyyy'),
      TRY_TO_DATE(Claim_Date, 'dd/MM/yyyy'),
      DATE('1900-01-01')
    )
  END AS Claim_Date,

  COALESCE(
    TRY_CAST(NULLIF(REGEXP_REPLACE(Claim_Amount, '[R,\\s]', ''), '') AS DOUBLE),
    0
  ) AS Claim_Amount,

  COALESCE(NULLIF(TRIM(Claim_Status), ''), 'UNKNOWN') AS Claim_Status,
  COALESCE(NULLIF(TRIM(Fraud_Flag), ''), 'Unknown') AS Fraud_Flag
FROM insurance_data_clean;

SELECT *
FROM insurance_data_clean
```

See performance (2)

3. Fix inconsistent text

09:45 PM (3s) 17: Fix inconsistent text

```sql
CREATE OR REPLACE TABLE insurance_data_clean AS
SELECT
  Customer_ID,
  Age,

  -- Gender normalization
  CASE
    WHEN UPPER(TRIM(Gender)) IN ('M','MALE') THEN 'Male'
    WHEN UPPER(TRIM(Gender)) IN ('F','FEMALE') THEN 'Female'
    ELSE 'Unknown'
  END AS Gender,

  -- Province normalization
  CASE
    WHEN UPPER(TRIM(Province)) IN ('WC','WESTERN CAPE') THEN 'Western Cape'
    WHEN UPPER(TRIM(Province)) IN ('EC','EASTERN CAPE') THEN 'Eastern Cape'
    WHEN UPPER(TRIM(Province)) IN ('KZN','KWAZULU-NATAL','KWAZULU NATAL') THEN 'KwaZulu-Natal'
    WHEN UPPER(TRIM(Province)) IN ('FS','FREE STATE') THEN 'Free State'
    WHEN UPPER(TRIM(Province)) IN ('GP','GAUTENG') THEN 'Gauteng'
    WHEN UPPER(TRIM(Province)) IN ('NW','NORTH WEST','NORTHWEST') THEN 'North West'
    WHEN UPPER(TRIM(Province)) IN ('LP','LIMPOPO') THEN 'Limpopo'
    WHEN UPPER(TRIM(Province)) IN ('MP','MPUMALANGA') THEN 'Mpumalanga'
    ELSE 'Unknown'
  END AS Province,

  Monthly_Income,
  Join_Date,
  Policy_ID,

  -- Policy_Type normalization (FIXED: added Funeral and Auto variants)
  CASE
    WHEN UPPER(TRIM(Policy_Type)) IN ('LIFE','L') THEN 'Life'
    WHEN UPPER(TRIM(Policy_Type)) IN ('MOTOR','CAR','AUTO') THEN 'Motor'
    WHEN UPPER(TRIM(Policy_Type)) IN ('HOME','HOUSE','PROPERTY') THEN 'Home'
    WHEN UPPER(TRIM(Policy_Type)) IN ('HEALTH','MEDICAL') THEN 'Health'
    WHEN UPPER(TRIM(Policy_Type)) IN ('FUNERAL','F') THEN 'Funeral'
    ELSE 'Unknown'
  END AS Policy_Type,
```

09:45 PM (3s) 17: Fix inconsistent text

```sql
  END AS Policy_Type,

  Premium_Amount,

  -- Policy_Status normalization
  CASE
    WHEN UPPER(TRIM(Policy_Status)) IN ('ACTIVE','A') THEN 'ACTIVE'
    WHEN UPPER(TRIM(Policy_Status)) IN ('CANCELLED','CANCELED','C') THEN 'CANCELLED'
    WHEN UPPER(TRIM(Policy_Status)) IN ('LAPSED','L') THEN 'LAPSED'
    WHEN UPPER(TRIM(Policy_Status)) IN ('DORMANT','D') THEN 'DORMANT'
    ELSE 'UNKNOWN'
  END AS Policy_Status,

  Claim_ID,
  Claim_Date,
  Claim_Amount,

  -- Claim_Status normalization
  CASE
    WHEN UPPER(TRIM(Claim_Status)) IN ('APPROVED','A','IN REVIEW') THEN 'APPROVED'
    WHEN UPPER(TRIM(Claim_Status)) IN ('REJECTED','DENIED','R','DECLINED') THEN 'REJECTED'
    WHEN UPPER(TRIM(Claim_Status)) IN ('PENDING','P') THEN 'PENDING'
    ELSE 'UNKNOWN'
  END AS Claim_Status,

  -- Fraud_Flag normalization
  CASE
    WHEN UPPER(TRIM(Fraud_Flag)) IN ('Y','YES','1','TRUE','FRAUD') THEN 'Fraud'
    WHEN UPPER(TRIM(Fraud_Flag)) IN ('N','NO','0','FALSE','NON-FRAUD') THEN 'Non-Fraud'
    ELSE 'Unknown'
  END AS Fraud_Flag

FROM insurance_data_clean;

SELECT *
FROM insurance_data_clean
```

> See performance (2)

4. Ensure correct data types

09:44 PM (2s) 14: correct data types:

```sql
CREATE OR REPLACE TABLE insurance_data_clean AS
SELECT
  Customer_ID,
  Age,
  Gender,
  Province,
  TRY_CAST(NULLIF(REGEXP_REPLACE(Monthly_Income, '[R,\\s]', ''), '') AS DOUBLE) AS Monthly_Income,
  COALESCE(
    TRY_TO_DATE(Join_Date, 'yyyy-MM-dd'),
    TRY_TO_DATE(Join_Date, 'dd-MMM-yyyy'),
    TRY_TO_DATE(Join_Date, 'MM/dd/yyyy'),
    TRY_TO_DATE(Join_Date, 'dd/MM/yyyy')
  ) AS Join_Date,
  Policy_ID,
  Policy_Type,
  TRY_CAST(NULLIF(REGEXP_REPLACE(Premium_Amount, '[R,\\s]', ''), '') AS DOUBLE) AS Premium_Amount,
  Policy_Status,
  Claim_ID,
  COALESCE(
    TRY_TO_DATE(Claim_Date, 'yyyy-MM-dd'),
    TRY_TO_DATE(Claim_Date, 'dd-MMM-yyyy'),
    TRY_TO_DATE(Claim_Date, 'MM/dd/yyyy'),
    TRY_TO_DATE(Claim_Date, 'dd/MM/yyyy')
  ) AS Claim_Date,
  TRY_CAST(NULLIF(REGEXP_REPLACE(Claim_Amount, '[R,\\s]', ''), '') AS DOUBLE) AS Claim_Amount,
  Claim_Status,
  Fraud_Flag

FROM workspace.dataforge_work.insurance_data
```

See performance (1) Optimize

5. Remove invalid records:

09:46 PM (3s) 18: Remove invalid records

```sql
CREATE OR REPLACE TABLE insurance_data_clean AS
SELECT *
FROM insurance_data_clean
WHERE
    -- IDs must not be Unknown
    Customer_ID <> 'Unknown'
    AND Policy_ID <> 'Unknown'
    AND Claim_ID <> 'Unknown'

    -- Age must be realistic
    AND Age BETWEEN 18 AND 100

    -- Gender must be valid
    AND Gender IN ('Male','Female','Unknown')

    -- Province must be one of the 9 SA provinces or Unknown
    AND Province IN (
        'Western Cape','Eastern Cape','KwaZulu-Natal','Free State',
        'Gauteng','North West','Limpopo','Mpumalanga','Unknown'
    )

    -- Monthly_Income, Premium_Amount, Claim_Amount must be non-negative
    AND Monthly_Income >= 0
    AND Premium_Amount >= 0
    AND Claim_Amount >= 0

    -- Dates must not be placeholder
    AND Join_Date <> DATE('1900-01-01')
    AND Claim_Date <> DATE('1900-01-01')

    -- Policy_Status must be valid
    AND Policy_Status IN ('ACTIVE','CANCELLED','LAPSED','UNKNOWN')

    -- Policy_Type must be valid (FIXED: added Funeral)
    AND Policy_Type IN ('Life','Motor','Home','Health','Funeral','Unknown')

    -- Claim_Status must be valid
    AND Claim_Status IN ('APPROVED','REJECTED','PENDING','UNKNOWN')
```


```sql
    -- Claim_Status must be valid
    AND Claim_Status IN ('APPROVED','REJECTED','PENDING','UNKNOWN')
```


```sql
    -- Fraud_Flag must be valid
    AND Fraud_Flag IN ('Fraud','Non-Fraud','Unknown')
```

See performance (1) Optimize

6. Final table preview

09:46 PM (1s) 19

```
SELECT *
FROM insurance_data_clean
```

See performance (1) Optimize

Table +

	Customer_ID	Age	Gender	Province	Monthly_Income	Join_Date	Policy_ID	Policy_Type	Premium_Amount	Policy_S
1	CUST1000	44	Male	Gauteng	26064	2020-05-30	POL5000	Home	1495	ACTIVE
2	CUST1001	41	Male	Free State	3500	2025-06-16	POL5001	Motor	746.28	ACTIVE
3	CUST1002	33	Female	Limpopo	29552.92	2020-06-14	POL5002	Health	2325.42	ACTIVE
4	CUST1003	41	Female	Western Cape	3500	2024-06-10	POL5003	Motor	885.13	ACTIVE
5	CUST1005	33	Female	Limpopo	0	2024-04-13	POL5005	Home	2210.31	ACTIVE
6	CUST1008	42	Male	North West	31654.77	2022-03-28	POL5008	Home	1410.71	ACTIVE
7	CUST1009	41	Male	Limpopo	26957.76	2020-11-02	POL5009	Funeral	326.92	ACTIVE
8	CUST1010	47	Female	Western Cape	15268.58	2021-05-18	POL5010	Motor	1840.98	ACTIVE
9	CUST1012	41	Male	Western Cape	38945.52	2021-08-24	POL5012	Home	1734	ACTIVE
10	CUST1013	59	Female	Gauteng	34633.66	2021-01-27	POL5013	Life	706.33	ACTIVE
11	CUST1014	36	Female	Free State	17447.69	2021-11-06	POL5014	Motor	1038.67	ACTIVE
12	CUST1015	45	Unknown	Mpumalanga	15999.79	2024-01-23	POL5015	Home	1921	ACTIVE
13	CUST1016	41	Male	North West	10559.66	2023-09-06	POL5016	Health	2331.19	ACTIVE
14	CUST1017	23	Male	KwaZulu-Natal	14870.45	2024-09-15	POL5017	Home	3333.73	LAPSED
15										

426 rows | 0.70s runtime Refreshed 21 minutes ago